Pose-Invariant Face Recognition Using Markov Random Fields

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***Abstract*— One of the key challenges for current face recognition techniques is how to handle pose variations between the probe and gallery face images. In this paper, we present a method for reconstructing the virtual frontal view from a given nonfrontal face image using Markov random fields (MRFs) and an efficient variant of the belief propagation algorithm. In the proposed approach, the input face image is divided into a grid of overlapping patches, and a globally optimal set of local warps is estimated to synthesize the patches at the frontal view. A set of possible warps for each patch is obtained by aligning it with images from a training database of frontal faces. The alignments are performed efficiently in the Fourier domain using an extension of the Lucas–Kanade algorithm that can handle illumination variations. The problem of finding the optimal warps is then formulated as a discrete labeling problem using an MRF. The reconstructed frontal face image can then be used with any face recognition technique. The two main advantages of our method are that it does not require manually selected facial landmarks or head pose estimation. In order to improve the performance of our pose normalization method in face recognition, we also present an algorithm for classifying whether a given face image is at a frontal or nonfrontal pose. Experimental results on different datasets are presented to demonstrate the effectiveness of the proposed approach.**

***Index Terms*— Belief propagation, frontal face synthesizing, Markov random fields, pose-invariant face recognition.**

I. Introduction

FACE recognition has been one of the most active research topics in computer vision and pattern recognition for more than two decades. The applications of face recognition can be found in telecommunication, law enforcement, biometrics and surveillance. Although there have been some early successes in automatic face recognition, it is still far from being completely solved, especially in uncontrolled environments. In fact, the performance of most of current face recognition systems drops significantly when there are variations in pose, illumination and expression [1].

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Pose variations can be considered as one of the most important and challenging problems in face recognition. As the viewpoint varies, the 2D facial appearance will change because the human head has a complex nonplanar geometry. Magnitudes of variations of innate characteristics, which distinguish one face from another, are often smaller than magnitudes of image variations caused by pose variations [2]. Popular frontal face recognition algorithms, such as Eigenfaces [3] or Fisherfaces [4], [5], usually have low recognition rates under pose changes as they do not take into account the 3D alignment issue when creating the feature vectors for matching.

Existing methods for face recognition across pose can be roughly divided into two broad categories: (1) techniques that rely on 3D models, and (2) 2D techniques. In the first type of approaches, the morphable model proposed by Blanz and Vetter [6] fits a 3D model to an input face using the prior knowledge of human faces and image-based reconstruction. The main drawback of this algorithm is that it requires many manually selected landmarks for initialization. Furthermore, the optimization process is computationally expensive and often converges to local minima due to a large number of parameters that need to be determined. Another recently proposed method by Biswas and Chellappa [7] estimates the facial albedo and pose at the same time using a stochastic filtering framework and performs recognition on the reconstructed frontal faces. The disadvantage of this approach lies in the use of an iterative algorithm for updating the albedo and pose estimates leading to accumulation of errors from step to step. Given a nonfrontal face image, the 3D pose normalization algorithm proposed by Asthana *et al.* [8] uses the pose-dependent correspondences between 2D landmark points and 3D model vertices in order to synthesize the frontal view. The main drawback of this method is the dependence on the fitting of landmarks using the Active Appearance Model (AAM) [9].

On the other hand, 2D techniques do not require the 3D prior information for performing pose-invariant face recognition. The AAM algorithm proposed by Cootes *et al.* [9] fits a statistical appearance model to the input image by learning the relationship between perturbations in the model parameters and the induced image errors. The main disadvantage of this approach is that each training image requires a large number of manually annotated landmarks. Gross *et al.* [10] proposed the eigen light-field (ELF) method that unifies all possible appearances of faces in different poses within a 4D space (two viewing directions and two pixel positions). However, this method discards shape variations due to different identity as it requires a restricted alignment of the image to the

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light field space. Recently, Prince *et al.* [11] use an affine mapping and pose information to generate the observation space from the identity space. In the approach proposed by Castillo and Jacobs [12], the cost of stereo matching was used in face recognition across pose without performing 3D reconstruction. Sarfraz and Hellwich [13] try to solve the problem by modeling the joint appearance of gallery and probe images across pose in a Bayesian framework.

Patch-based approaches for face recognition under varying poses have received significant attention from the research community. These methods were motivated by the fact that a 3D face is composed of many planar local surfaces and thus, an out-of-plane rotation, although nonlinear under 2D imaging projection, can be approximated by linear transformations of 2D image patches. As a result, modeling a face as a collection of subregions/patches is more robust to pose variations than the holistic appearance. In the method proposed by Kanade and Yamada [14], each patch has a utility score based on pixel differences, and the recognition is performed using a Gaussian probabilistic model and a Bayesian classifier. Ashraf *et al.* [15] extended this approach by learning the patch correspondences based on 2D affine transforms. The problem with these approaches is that the transformations are optimized locally without taking into account the global consistency of the patches. In [16], linear regressions are performed on local patches in order to synthesize the virtual frontal view. Another approach proposed by [17] measures the similarities of local patches by correlations in a subspace constructed by Canonical Correlation Analysis (CCA). However, the common drawback of these two algorithms is that the head pose of the input face image needs to be known a priori. Arashloo and Kittler [18] present a method for estimating the deformation parameters of local patches using Markov Random Fields (MRFs). The disadvantage of this approach is that it depends on estimating the global geometric transformation between the template and the target images. Although designed specifically for handling expression variations in face recognition, another related work is the method proposed by Liao and Chung [19], which formulates the face recognition problem as a deformable image registration problem using MRFs. However, this approach also depends on the extraction of salient regions from the face images.

In this paper, a patch-based method for synthesizing the virtual frontal view from a given nonfrontal face image using MRFs and an efficient variant of the BP algorithm is proposed. By aligning each patch in the input image with images from a training database of frontal faces, a set of possible warps is obtained for that patch. The alignments are then carried out efficiently using an illumination invariant extension of the Lucas–Kanade (LK) algorithm [20] in the frequency domain. The objective of the algorithm is to find the globally optimal set of local warps that can be used to predict the image patches at the frontal view. This goal is achieved by considering the problem as a discrete labeling problem using an MRF. In our approach, the cost functions of the MRF are not just the simple sum of squared differences (SSD) between patches but are modified to reduce the effect of illumination variations. The optimal labels are obtained using a variant of the BP algorithm with *message scheduling* and *dynamic label pruning* [21]. The two main advantages of our approach over other state-of-the-art algorithms are that: (1) it does not require manually selected landmarks, and (2) no global geometric transformation is needed. Furthermore, we also present a method that is able to classify whether an input face image is frontal or nonfrontal. This method extracts dense SIFT descriptors [22] from the input face image and performs classification using the Support Vector Machine (SVM) algorithm [23]. It is used to improve the performance of our pose normalization technique in face recognition. Experimental results on the FERET [24], CMU-PIE [25] and Multi-PIE [26] databases are presented to demonstrate the effectiveness of the proposed algorithm.

The remainder of this paper is organized as follows. Section II describes the illumination-insensitive alignment method based on the LK algorithm. The reconstruction of the virtual frontal view using MRFs and BP is discussed in Section III. The frontal-view classification algorithm is presented in Section IV. Next, in Section V, we show the experimental results in both frontal face reconstruction and pose-invariant face recognition. Section VI concludes this paper.

II. Illumination Insensitive Patch Alignment

A. Alignment of Local Patches Using Weighted Lucas–Kanade

Assume that we have two images, the probe image I and the gallery image T, captured at two different viewpoints. The images are divided into M blocks (rectangle patches) and for each pair of corresponding patches, I_i and T_i, a local warp W_i is estimated to align them using the weighted Lucas–Kanade (LK) algorithm [27]. The warp W_i, parameterized by the vector $\mathbf{p}_i$, minimizes the following error function:

$$\begin{aligned} E_i(\mathbf{p}_i) &= ||I_i(\mathbf{p_i}) - T_i(\mathbf{0})||^2_{\mathbf{Q}} \\ &= [I_i(\mathbf{p}_i) - T_i(\mathbf{0})]^T \mathbf{Q}[I_i(\mathbf{p}_i) - T_i(\mathbf{0})] \end{aligned} \quad (1)$$

where $\mathbf{Q}$ is a symmetric, positive semi-definite weighting matrix. Note that $I_i(\mathbf{p_i})$ and $T_i(\mathbf{0})$ are both vectorized image patches. Equation (1) becomes the standard LK objective function [28] when $\mathbf{Q}$ is an identity matrix. If $W(\mathbf{p})$ is an affine warp with parameters $\mathbf{p} = (p_1, p_2, p_3, p_4, p_5, p_6)^T$, it can be written as

$$W(\mathbf{p}) = \begin{pmatrix} 1+p_1 & p_3 & p_5 \\ p_2 & 1+p_4 & p_6 \end{pmatrix}.$$

The transformed image patch $I_i(\mathbf{p_i})$ is obtained by applying the warp to all the pixels in I_i.

Equation (1) is highly nonlinear and thus, can be linearized by performing the first order Taylor expansion on $I_i(\mathbf{p}_i + \Delta\mathbf{p}_i)$

$$E_i(\mathbf{p}_i) \approx ||I_i(\mathbf{p}_i) + \mathbf{J}_i \Delta\mathbf{p}_i - T_i(\mathbf{0})||^2_{\mathbf{Q}} \quad (2)$$

where $\mathbf{J}_i = \left(\frac{\partial I_i(\mathbf{p}_i)}{\partial \mathbf{p}_i}\right)^T$ is the Jacobian of $I_i(\mathbf{p}_i)$. The value of $\Delta\mathbf{p}_i$ that minimizes (2) is given by

$$\Delta\mathbf{p}_i = \mathbf{H}_i^{-1}\mathbf{J}_i^T\mathbf{Q}[T_i(\mathbf{0}) - I_i(\mathbf{p}_i)] \quad (3)$$

where the pseudo-Hessian matrix is defined as

$$\mathbf{H}_i = \mathbf{J}_i^T\mathbf{Q}\mathbf{J}_i = \frac{\partial I_i(\mathbf{p}_i)}{\partial \mathbf{p}_i}\mathbf{Q}\frac{\partial I_i(\mathbf{p}_i)}{\partial \mathbf{p}_i}^T. \quad (4)$$

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An iterative solution to (1) can be obtained by iteratively solving for $\Delta\mathbf{p}_i$ and updating the warp parameters $\mathbf{p}_i = \mathbf{p}_i + \Delta\mathbf{p}_i$ until convergence.

B. Illumination Insensitive Alignment Based on Gabor Features

It is known that the original LK algorithm is very sensitive to changes in illumination [29]. The main advantage of the weighted LK algorithm over the original method is that illumination variations can be handled by encoding the prior knowledge of the correlation and salience of image pixels into $\mathbf{Q}$. As a result, choosing an appropriate weighting matrix $\mathbf{Q}$ is an important problem with the weighted LK algorithm. In a recently proposed method [20], it is shown that robustness against illumination changes as well as low computational complexity can be achieved by constructing $\mathbf{Q}$ from the Fourier transforms of a bank of Gabor filters [30].

A 2-D Gabor filter $g_{\mu,\nu}(z)$ where $z = (x, y)$ is defined as the product of an elliptical Gaussian envelope and a complex plane wave [30]

$$g_{\mu,\nu}(z) = \frac{||k_{\mu,\nu}||^2}{\sigma^2} e^{-\frac{||k_{\mu,\nu}||^2||z||^2}{2\sigma^2}} \left[e^{ik_{\mu,\nu}\cdot z} - e^{-\frac{\sigma^2}{2}} \right] \quad (5)$$

where ν and μ denote the scale and orientation of the Gabor filter, respectively. σ is the parameter determining the ratio of the Gaussian window width to the wavelength. The wave vector $k_{\mu,\nu}$ is defined as

$$k_{\mu,\nu} = k_\nu e^{i\phi_\mu} \quad (6)$$

where $k_\nu = \frac{k_{max}}{f^\nu}$ and $\phi_\mu = \frac{\pi\mu}{8}$. f is the spacing factor between kernels in the frequency domain and k_{max} is the maximum frequency. The term $e^{-\frac{\sigma^2}{2}}$ is subtracted in order to make the filter invariant to illumination changes. In this paper, a bank of 40 Gabor filters corresponding to five different scales, $\nu = 0, \ldots, 4$, and eight orientations, $\mu = 0, \ldots, 7$, was used in the experiments. The values of other parameters were set as follow: $\sigma = 2\pi$, $k_{max} = \frac{\pi}{2}$ and $f = \sqrt{2}$ [30].

Assume that $\mathbf{g}_k$ $(k = 1, \ldots, K)$ is the k-th impulse response of a bank of K Gabor filters, the alignment error can be written as the sum of squared differences (SSD) across all filter responses of the warped probe patch and the gallery patch

$$E_i(\mathbf{p}_i) = ||\{\mathbf{g}_k * I_i(\mathbf{p}_i)\}_{k=1}^K - \{\mathbf{g}_k * T_i(\mathbf{0})\}_{k=1}^K||^2 \quad (7)$$

where $\{.\}_{k=1}^K$ denotes the concatenation operation, i.e. $\{\mathbf{x}_k\}_{k=1}^K = [\mathbf{x}_1^T, \ldots, \mathbf{x}_K^T]^T$, and $*$ represents the 2D convolution operation. Using Parseval's relation [31], the error in (7) can be estimated in the Fourier domain as

$$E_i(\mathbf{p}_i) = ||\hat{I}_i(\mathbf{p}_i) - \hat{T}_i(\mathbf{0})||_{\mathbf{S}}^2 \quad (8)$$

where $\mathbf{S} = \sum_{k=1}^K (\text{diag}(\hat{\mathbf{g}}_k))^T \text{diag}(\hat{\mathbf{g}}_k)$ and $\hat{I}_i$, $\hat{T}_i$, $\hat{\mathbf{g}}_k$ are the 2D Fourier transforms of I_i, T_i, $\mathbf{g}_k$, respectively. It is worth noting that $\mathbf{S}$ is a diagonal matrix and can be precomputed. As the 2D Fourier transform of a signal of length L is computed by pre-multiplying it by the $L \times L$ Fourier matrix $\mathbf{F}$, (8) is equivalent to

$$E_i(\mathbf{p}_i) = ||I_i(\mathbf{p}_i) - T_i(\mathbf{0})||_{\mathbf{F}^T\mathbf{S}\mathbf{F}}^2. \quad (9)$$

From (3), the update $\Delta\mathbf{p}_i$ is obtained as

$$\Delta\mathbf{p}_i = \mathbf{H}_{flk}^{-1}(\mathbf{F}\mathbf{J}_i)^T\mathbf{S}\mathbf{F}[T_i(\mathbf{0}) - I_i(\mathbf{p}_i)] \quad (10)$$

where $\mathbf{H}_{flk} = (\mathbf{F}\mathbf{J}_i)^T\mathbf{S}(\mathbf{F}\mathbf{J}_i)$ is the pseudo-Hessian. In order to perform the update efficiently, the FFT algorithm [31] is applied to estimate the Fourier transforms of the columns of the Jacobian matrix $\mathbf{J}$ and the error image $T_i(\mathbf{0}) - I_i(\mathbf{p}_i)$ at each iteration.

The above formulation of the LK algorithm is known as the forward additive (FA) algorithm. In order to improve the computational efficiency, an extension to the forward additive LK called the inverse compositional (IC) algorithm was proposed in [32]. In this approach, the error function is formulated by linearizing $T_i(\Delta\mathbf{p})$ rather than $I_i(\mathbf{p}_i + \Delta\mathbf{p}_i)$

$$E \approx ||T_i(\mathbf{0}) + \mathbf{J}_{i(ic)}\Delta\mathbf{p}_i - I_i(\mathbf{p}_i)|| \quad (11)$$

where $\mathbf{J}_{i(ic)} = \left(\frac{\partial T_i(\mathbf{0})}{\partial \mathbf{p}_i}\right)^T$. The update $\Delta\mathbf{p}_i$ is can be solved as [20]

$$\Delta\mathbf{p}_i = \mathbf{B}[I_i(\mathbf{p}_i) - T_i(\mathbf{0})] \quad (12)$$

where $\mathbf{B} = \mathbf{H}_{flk(ic)}^{-1}(\mathbf{F}\mathbf{J}_{i(ic)})^T\mathbf{S}\mathbf{F}$. Note that the pseudo-Hessian $\mathbf{H}_{flk(ic)} = \mathbf{J}_{i(ic)}^T\mathbf{F}^T\mathbf{S}\mathbf{F}\mathbf{J}_{i(ic)}$ is computed only once for all iterations.

If N and n are the number of pixels in an image patch and the number of warp parameters, respectively, the computational complexity of the inverse compositional algorithm is $O(n^2 + nN)$ per iteration [20]. This is significantly better than the case of the forward additive approach where the computational complexity is $O(n^3 + n^2N + nN\log N)$ per iteration.

III. Frontal Face Reconstruction Using Markov Random Fields

Given an input image I of a nonfrontal face and M training face images $T^{(k)}$, $k = 1, \ldots, M$ captured at the frontal pose, all of them are divided into the same regular grid of N overlapping patches of size $w \times h$. A set of M possible local warps $\mathcal{P}_i = \{\mathbf{p}_i^{(k)} : k = 1, \ldots, M\}$ can be estimated for each patch I_i, by aligning it with the corresponding patches of the training images using the method presented in Section II-B. By aligning the patches in the nonfrontal views with the ones in the frontal views, we can obtain the information about how the local patches are transformed as a result of the 3D rotation of the face. The goal of our algorithm is to find a globally optimal set of warps for all the patches in the input image such that we can predict the input face at the frontal pose by transforming these patches using the obtained warps. This problem can be turned into a discrete labeling problem with a well defined objective function using a discrete MRF. Note that in our approach, the training database need not contain the frontal images of the person in the input image I.

A. Markov Random Fields

In the proposed algorithm, lattice points whose local patches are inside the image form a set of MRF nodes $\mathcal{V}$ (Figure 1). The set of warps $\mathcal{P}_i$ can be considered as the set of possible

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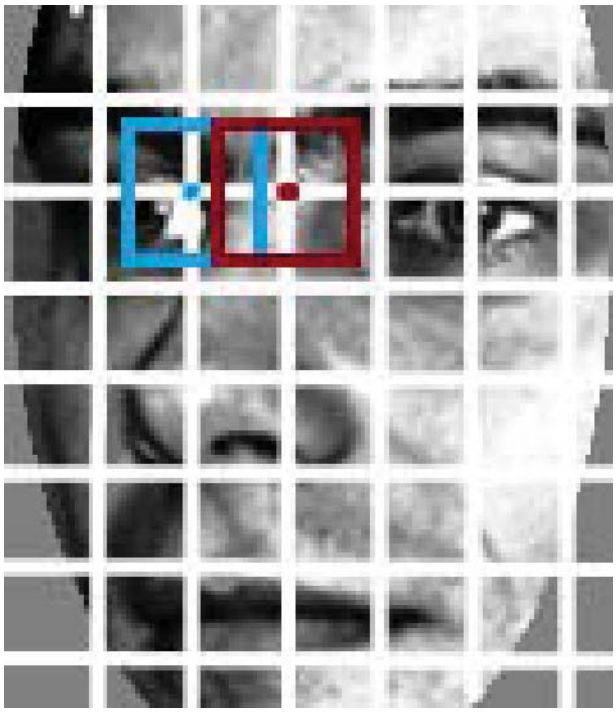


Fig. 1. Two neighboring MRF nodes with overlapping patches.

labels for node i. A 4-connected neighborhood system is then created by edges $\mathcal{E}$ of the MRF.

The *single node potential* $E_i(\mathbf{p}_i)$ penalizes the cost of assigning the warp $\mathbf{p}_i^{(k)}$ to node i. It can be defined using (9) as

$$E_i(\mathbf{p}_i) = ||I_i(\mathbf{p}_i) - T_i^{(k)}(\mathbf{0})||^2_{\mathbf{F}^T\mathbf{SF}} \tag{13}$$

where $\mathbf{p}_i \in \mathcal{P}_i$ and k is the index of the training image that corresponds to the warp $\mathbf{p}_i$. The *pairwise potential* $E_{ij}(\mathbf{p}_i, \mathbf{p}_j)$ is the cost of label discrepancy between two neighboring nodes i and j. In other words, this smoothness term measures how well neighboring labels agree at the region of overlap. In order to reduce the effect of illumination changes, the local patches are normalized by subtracting the means and dividing by the standard deviations before estimating the sum of squared difference in the overlapping region. $E_{ij}(\mathbf{p}_i, \mathbf{p}_j)$ can be written as

$$E_{ij}(\mathbf{p}_i, \mathbf{p}_j) = \sum_{\mathbf{x} \in \text{node } i \cap \text{node } j} \left(\hat{I}_i(\mathbf{x}; \mathbf{p}_i) - \hat{I}_j(\mathbf{x}; \mathbf{p}_j)\right)^2 \tag{14}$$

where $\mathbf{p}_i \in \mathcal{P}_i$, $\mathbf{p}_j \in \mathcal{P}_j$ and $\hat{I}_i(\mathbf{x}; \mathbf{p}_i)$ denotes the intensity value at the location $\mathbf{x}$ in $\hat{I}_i(\mathbf{p}_i)$. The intensity at $\mathbf{x}$ of the normalized patch $\hat{I}_i(\mathbf{p}_i)$ is obtained as

$$\hat{I}_i(\mathbf{x}; \mathbf{p}_i) = \frac{I_i(\mathbf{x}; \mathbf{p}_i) - \mu_i}{\sigma_i} \tag{15}$$

where μ_i and σ_i are the mean and standard deviation of the intensities, respectively, of the local patch I_i without applying any warping function. As local deformations do not affect the intensities of image pixels, the values of μ_i and σ_i can be precomputed to improve the speed of the algorithm. The optimal labeling or the optimal set of warps $\{\hat{\mathbf{p}}_i\}_{i=1}^M$ can be found by minimizing the following energy function:

$$E(\{\mathbf{p}_i\}_{i=1}^M) = \sum_{i \in \mathcal{V}} E_i(\mathbf{p}_i) + \lambda \sum_{(i,j) \in \mathcal{E}} E_{ij}(\mathbf{p}_i, \mathbf{p}_j) \tag{16}$$

where λ is a regularization parameter that controls the interaction between the single node potentials and pairwise potentials.

B. Priority Belief Propagation and Label Pruning

The minimization of (16) can be performed by using an optimization method for MRFs known as Belief Propagation (BP) [33]. It is an inference technique that works by passing local messages along the nodes of a MRF. In the case of Markov networks without loops, BP is an exact inference method. Even in networks with loops, it often leads to good approximate results [34]. Using negative logarithmic probabilities, a message from node i to node j at time t is defined as

$$m_{ij}^t(\mathbf{p}_i) = \min_{\mathbf{p}_i \in \mathcal{P}_i} \{E_i(\mathbf{p}_i) + \lambda E_{ij}(\mathbf{p}_i, \mathbf{p}_j) + \sum_{k:k \neq j, (k,i) \in \mathcal{E}} m_{ki}^{t-1}(\mathbf{p}_i)\}. \tag{17}$$

Assume that all messages converge after s iterations, the belief of node i for $\mathbf{p}_i \in \mathcal{P}_i$, $b_i(\mathbf{p}_i)$ is computed as

$$b_i(\mathbf{p}_i) = -E_i(\mathbf{p}_i) - \sum_{k:(k,i) \in \mathcal{E}} m_{ki}^s(\mathbf{p}_i). \tag{18}$$

The warp $\hat{\mathbf{p}}_i = \underset{\mathbf{p}_i \in \mathcal{P}_i}{\text{argmax}}\, b_i(\mathbf{p}_i)$ is selected as the optimal label for node i.

It is known that the standard BP is slow and requires many iterations to converge [35]. In [21], two extensions to the standard BP were proposed in order to improve the speed and make the algorithm converge after a small number of iterations.

The first extension to the standard BP is the use of *dynamic label pruning*. If the number of active labels for a node is greater than $L_{\max}$, a user specified constant, label pruning will be applied to the node. The labels of a visited node are traversed in the descending order of relative belief $b_i^{\text{rel}}(\mathbf{p}_i)$, where the relative belief is defined as $b_i^{\text{rel}}(\mathbf{p}_i) = b_i(\mathbf{p}_i) - b_i^{\max}$ and $b_i^{\max}$ is the maximum belief of node i. Those labels $\mathbf{p}_i \in \mathcal{P}_i$ with $b_i^{\text{rel}}(\mathbf{p}_i) > b_{prune}$ are selected as active labels for node i. b_{prune} is the label pruning threshold belief. Furthermore, a label is declared as active only if it is not too similar to any of the already active labels in order to avoid choosing many similar labels and wasting a large part of the active label set. Two labels are considered similar if their normalized cross correlation is greater than a threshold T_{similar}. Note that a minimum number of labels $L_{\min}$ is always kept for each node. The complexity of updating the messages is reduced from $O(|L|^2)$ to $O(|L_{\max}|^2)$ by applying label pruning to BP [21]. In addition, the speed of BP can also be improved by precomputing the reduced matrices of pairwise potentials.

The second improvement is the use of *message scheduling* to determine the transmitting order for a node based on the confidence of that node about its labels. *The node most*

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confident about its label should be the first one to transmit outgoing messages to its neighbors [21]. The priority of a node is defined as $priority(i) = \frac{1}{|Q_i|}$ where $|Q_i|$ is the cardinality of the set $Q_i = \{\mathbf{p}_i \in \mathcal{P}_i : b_i^{\text{rel}}(\mathbf{p}_i) \geq b_{\text{conf}}\}$. b_{conf} is the confidence threshold belief. By employing this message scheduling in BP, the node that has the most informative messages will transmit first in order to increase the confidence of its neighbors. This helps the algorithm to converge only after only a small, fixed number of iterations. Furthermore, message scheduling also makes the neighbors of the transmitting node more tolerant to label pruning.

C. Frontal Face Recontruction

After running the priority belief propagation algorithm with label pruning, an optimal set of local warps $\{\hat{\mathbf{p}}_i\}_{i=1}^{N}$ is obtained. In order to synthesize the virtual frontal view, each patch I_i is transformed using the warp $\hat{\mathbf{p}}_i$ and all the transformed patches are combined together to create a frontal face image. As mentioned above, patches are sampled with some amount of overlap in order to reduce the blocking effect. In our approach, the intensity of a pixel in the overlapping region is computed as the average of the intensities of the overlapped patches in the same position. However, local patches can be better blended by using a similar approach to [36]. The Poisson solver discussed in Agrawal *et al.* [37] can be used to remove the intensity gradients along the seam created by two overlapping patches [38]. Finally, missing regions in the reconstructed frontal face image are filled using the facial symmetry constraint.

IV. Frontal-View Classification

In order to avoid degrading performance when applying the proposed pose compensation technique to face recognition, it is important to be able to automatically decide if the input face image is frontal or nonfrontal. In our approach, the frontal-view classification is performed using the Support Vector Machine (SVM) algorithm [23]. First, dense Scale Invariant Feature Transform (SIFT) descriptors are extracted from image grid points in order to obtain a representation that is robust to noise and illumination variations. The dimension of the concatenated descriptor vector is reduced for efficient processing by using Random Projection (RP). Finally, an SVM is employed to decide whether the face image is at the frontal pose or not. More details about SVM can be found in [39].

A. Dense SIFT Descriptors

One of the most popular methods for extracting keypoints from an image is the SIFT algorithm proposed by Lowe [22]. In this algorithm, a local descriptor is created at each detected keypoint by forming a histogram of gradient orientations and magnitudes of image pixels in a small window. The size of the local window is usually chosen at 16×16. It is then divided into sixteen 4×4 sub-windows. Gradient orientations and magnitudes are estimated within each sub-window and put into an 8 bin histogram. The histograms of the sub-windows are concatenated to create a 128-dimensional feature vector (descriptor) of the keypoint.

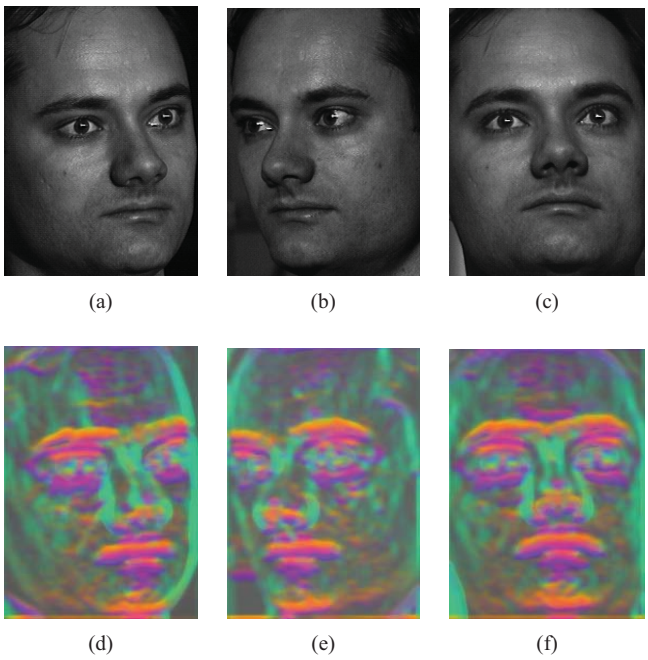


Fig. 2. (a)–(c) Input face images at different poses. (d)–(f) The corresponding visualizations of their dense SIFT descriptors. The first component of the descriptor is mapped into R + G + B, the second and third components are mapped into R − G and R/2 + G/2 − B, respectively.

In order to form a dense description of the input face image, local SIFT descriptors are extracted at regular image grid points, rather than only at keypoints, in the proposed approach. The advantage of this representation is that it does not depend on the matching of keypoints, which is often challenging when significant pose and illumination variations are present between the input images. This dense representation was also employed successfully for image alignment, gender classification and head-pose estimation in [40], [41], and [42], respectively. Figure 2 shows the input face images at different poses and their corresponding dense SIFT descriptors. In the second row of the figure, the first three principal components of each descriptor are mapped onto the principal components of the RGB color space in order to visualize purpose. Similar to [40], the first component is mapped into R + G + B, the second and third components are mapped into R − G and R/2 + G/2 − B, respectively.

B. Dimension Reduction Using Random Projection (RP)

As the dimension of the concatenated feature vector for the whole input face image is significantly large, techniques such as Principal Component Analysis (PCA) [43] can be used to project the concatenated feature vector into a lower-dimensional subspace. However, due to the large dimension of the feature space, the eigenvalue decomposition of the data covariance matrix will be very computationally expensive. A more efficient way to reduce the dimension of the feature vectors is by projecting them onto a random lower-dimensional subspace.

The main idea of random projection comes from the Johnson–Lindenstrauss (JL) lemma [44].

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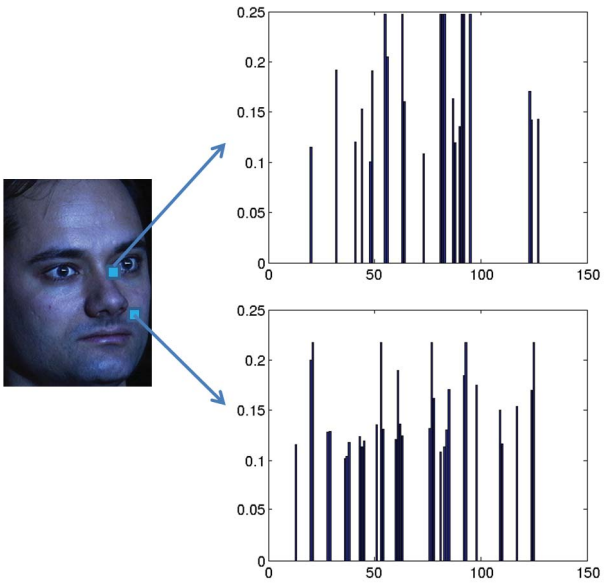


Fig. 3. SIFT descriptors extracted from two different locations in a face image.

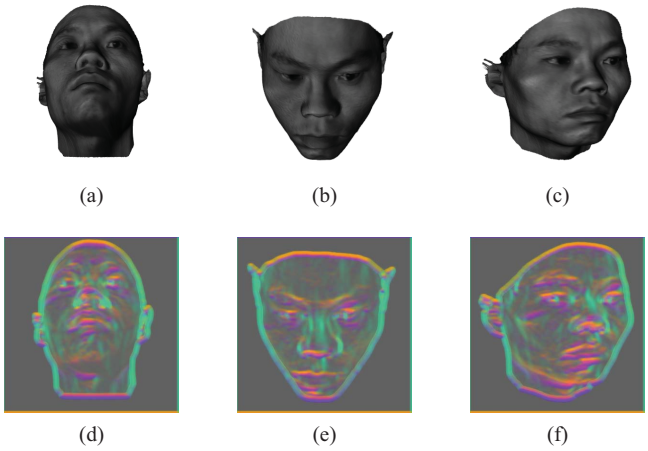


Fig. 4. (a)–(c) 3-D face model of a person in the USF 3-D database at different viewing angles. (d)–(f) Visualization of the corresponding dense SIFT descriptors.

Lemma 1: (Johnson–Lindenstrauss) Let $\epsilon \in (0, 1)$ be given. For every set Q of $\#(Q)$ points in $\mathcal{R}^N$, if n is a positive integer such that $n > n_0 = O\left(\frac{ln(\#(Q))}{\epsilon^2}\right)$, there exists a Lipschitz mapping $f : \mathcal{R}^N \rightarrow \mathcal{R}^n$ such that

$$(1-\epsilon)||\mathbf{u}-\mathbf{v}||^2 \leq ||f(\mathbf{u})-f(\mathbf{v})||^2 \leq (1+\epsilon)||\mathbf{u}-\mathbf{v}||^2 \quad (19)$$

for all $\mathbf{u}, \mathbf{v} \in Q$.

Basically, this lemma states that the pairwise distances between any two points are approximately maintained when the points are projected onto a random subspace of suitably high dimension. It is claimed in [45] that the performance of a wide variety of machine learning algorithms when given access to only randomly projected data is *essentially the same* as its performance on the original dataset.

In our implementation, each element $\phi_{i,j}$ of the random projection matrix Φ is generated independently according to the following simple distribution:

$$\phi_{i,j} = \sqrt{\frac{3}{n}} \begin{cases} +1 & \text{with probability } 1/6 \\ 0 & \text{with probability } 2/3 \\ -1 & \text{with probability } 1/6 \end{cases} \quad (20)$$

where n is the dimension of the random subspace. The mapping given by this matrix satisfies the JL lemma and is more computationally efficient compare to Gaussian distributed random matrices [46].

Because the majority of patches in a face image are uniform, when estimating the SIFT descriptors, there are many bins in the histogram of image gradients with zero values. As a result, the concatenated descriptor vector is sparse. Figure 3 shows the SIFT descriptors extracted from two different locations in a face image.

The sparsity of the concatenated SIFT descriptor vectors helps to further improve the efficiency of the random projection. For K-sparse signals (i.e., have at most K nonzero entries), the computational complexity of the random projection reduces from $O(nNC)$ to $O(nKC)$ for a dataset containing C vectors [46]. Furthermore, the embedding subspace dimension now depends only on the information content K of the dataset, not on its *cardinality* C as in the case of nonsparse signals. In other words, if the signals are K-sparse, the JL lemma holds for $n = O(K \log N)$ [45].

V. Experimental Results

A. Frontal-View Classification Using Dense SIFT Descriptors

The proposed frontal-view classification algorithm was trained using an SVM on 2D images generated from the 3D faces in the USF 3D database [47]. By rotating the 3D models and projecting them into the image plane, we can synthesize the 2D face images at different viewing angles. Face images with less than $\pm 5^\circ$ in both the yaw and pitch angles are labeled as frontal. Figure 4 shows the 2D face images of a person in the database generated at different poses and the visualization of their corresponding dense SIFT descriptors. As the USF 3D database contains the geometry as well as the texture information of the 3D faces, the face images at different illumination conditions can also be generated from the surface normals and albedo using the *Lambert's Cosine Law*

$$I_{i,j} = \rho_{i,j} \max(\mathbf{n}_{i,j}^T \mathbf{s}, 0) \quad (21)$$

where $\mathbf{s}$ is the direction of the light source, $I_{i,j}$, $\mathbf{n}_{i,j}$, and $\rho_{i,j}$ are the image intensity, surface normal and albedo at the pixel (i, j), respectively. This is necessary in order for the method to handle possible illumination variations in the test images.

We tested the proposed frontal-view classification algorithm on four different databases including the USF 3D database [47], FERET [24], CMU-PIE [25], and Multi-PIE [26]. For the USF 3D database, the synthesized face images were divided into five subsets. Four of them were used for training and the remaining subset was used for testing. It takes less than 4 seconds to perform the frontal-view classification for an input face image of size 130×150 on an Intel Xeon 2.13 GHz desktop.

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TABLE I
FRONTAL-VIEW CLASSIFICATION RATES FOR DIFFERENT DATASETS

Method	USF 3D Database	FERET	CMU-PIE	Multi-PIE
Dense SIFT + PCA	96.6%	95.7%	94.7%	94.3%
Our approach (Dense SIFT + RP)	**98.3%**	**97.2%**	**96.9%**	**94.9%**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

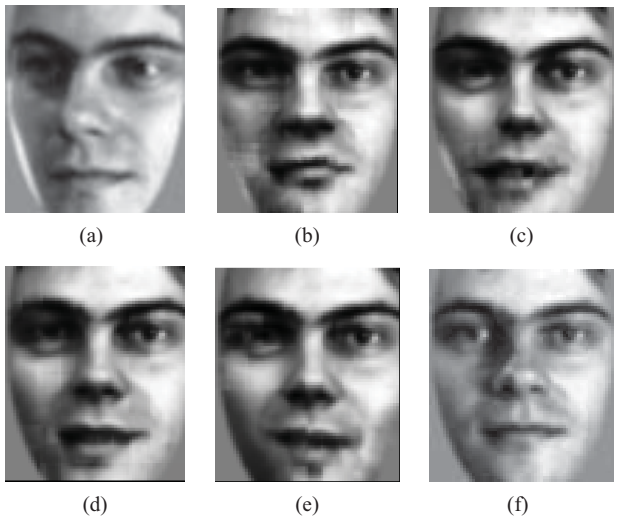
Fig. 5. Face images of a subject in the FERET database with varying viewpoints and illumination. (a) *ba*. (b) *bb*. (c) *bc*. (d) *bd*. (e) *be*. (f) *bf*. (g) *bg*. (h) *bh*. (i) *bi*. (j) *bk*.

Table I shows the classification rates for the four datasets. The results obtained using dense SIFT descriptors with PCA are also included for comparison. It can be seen from the table that, although the classification rates are high for both approaches, the one using dense SIFT and RP achieves better results.

B. Frontal Face Reconstruction

In this section, we present the results of reconstructing frontal views from nonfrontal face images using the proposed approach. Given an input face image, it is roughly aligned to the frontal faces in the training database using the coordinates of the two eyes. The input face and eye locations are detected automatically using the Viola–Jones object detection framework [48]. Similar to [8], different cascade classifiers are trained to locate the faces and eyes for the three rough pose classes (left half-profile, frontal and right half-profile). Each classifier can also handle pitch angles ranging from $-30°$ to $30°$. Positive training samples were cropped from the annotated face images of the first two hundred subjects of the Multi-PIE dataset [26] as well as from other datasets such as the USF 3D database [47], Pointing '04 [49], FacePix(30) [50] and LFW [51]. Negative samples were collected from a large number of random images on the Web. The input face image is translated, rotated and scaled so that the eyes map to canonical eye positions. Although this initial alignment process results in the difference in scale between the frontal and nonfrontal faces, the local warps of image patches are able to compensate for this variation, given the pose of the nonfrontal face is not too severe. Both the input and training images are smoothed by a 2D Gaussian filter in order to remove the noise as well as improve the accuracy of the estimation of image gradients in the alignment step.

The first dataset used in the experiments is the FERET dataset [24] that consists of images from two hundred subjects. Each subject in this database was captured at nine different view-points *ba*, *bb*, *bc*, *bd*, *be*, *bf*, *bg*, *bh*, *bi* which roughly correspond to nine viewing angles of 0°, 60°, 40°, 25°, 15°, −15°, −25°, −40°, −60°, respectively. The database also contains images denoted as *bk* which are frontal images corresponding to *ba*, but taken under different lighting.



(a) (b) (c) (d) (e) (f)

Fig. 6. Reconstructed frontal faces with various patch sizes. (a) Input. (b) 7×7. (c) 9×9. (d) 15×15. (e) 21×21. (f) Ground-truth.

Figure 5 shows different face images of a subject in the FERET database with varying pose and illumination.

One of the most important parameters in our method is the patch size. It should not be either too large or too small. If the patches are too small, they do not contain sufficient information for estimating the alignment parameters, especially when there are large displacements. A good patch size must provide enough overlapping in order to align corresponding patches between different views. On the other hand, if the patch size is too big, alignment parameters may not be estimated accurately [52] and blocking effects also appear. Figure 6 shows the reconstructed frontal faces from a nonfrontal face image using different patch sizes. It can be seen from the figure that if the patch size is too small or too large, there are many artifacts in the outputs. The patch size of 15×15 [Figure 6(d)] gave the best virtual frontal view when compared to the ground truth in [Figure 6(f)].

In all the experiments reported in this paper, the patch size was set at 15×15 and the gap between two neighboring MRF nodes was selected at ten pixels in order to have a sufficient amount of overlap between the neighboring patches.

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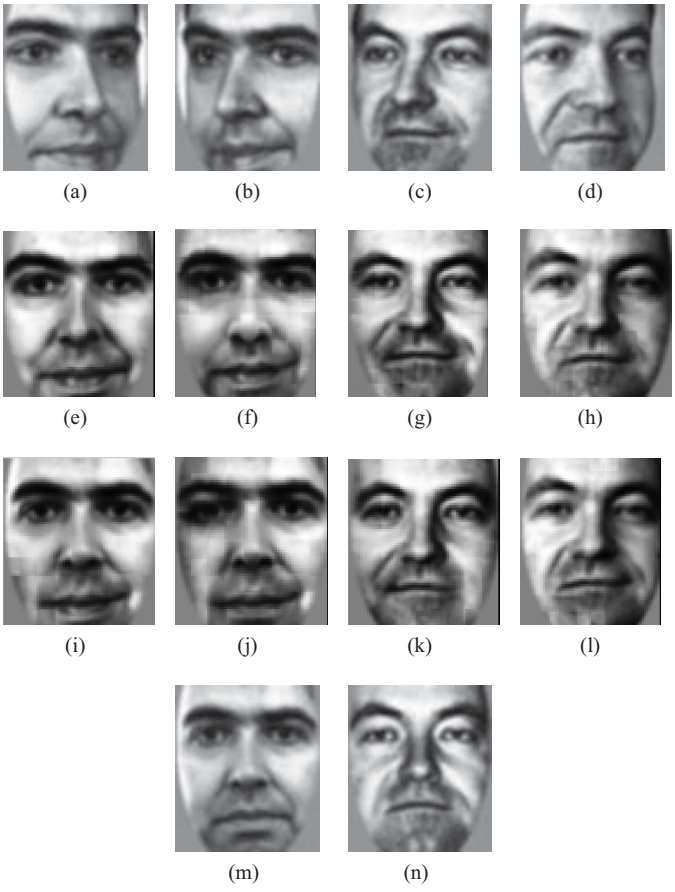


Fig. 7. Reconstructed frontal faces using training sets under different lighting. (a)–(d) Input images. (e)–(h) Results obtained using ba training set. (i)–(l) Results obtained using bk training set. (m) and (n) Ground truths.

$\lambda = 1$ was chosen to be the value of the regularization parameter. Two hundred frontal images denoted as *ba* from the FERET database were taken as the training set for guiding the alignment process. We observed that the number of iterations required for the priority BP algorithm with label pruning to converge is around five iterations. It takes less than two minutes to synthesize the frontal view for an input face image of size 130×150 on an Intel Xeon 2.13 GHz desktop.

In order to evaluate the performance of our approach in the case of varying illumination, another training set was formed from two hundred frontal face images of the FERET database taken under different lighting (those denoted as *bk*). The reconstructed frontal faces using two training sets *ba* and *bk* are shown in Figure 7. It can be seen from the figure that the difference in illumination between the input image and the training set does not affect the robustness of our algorithm. Both results obtained using *ba* and *bk* look very similar to each other and are close to the ground truths [Figure 7(m) and 7(n)].

The proposed algorithm was also tested on the CMU PIE database [25]. This database consists of face images taken from sixty eight subjects under thirteen different poses. The poses are denoted as $c05$ and $c29$ (the yawn angle about $\pm 22.5°$), $c37$ and $c11$ (the yawn angle about $\pm 45°$), and $c07$ and $c09$ (the pitch angle about $\pm 20°$). Figure 8 shows the synthesized frontal views of the same subject from the CMU PIE database at different poses. It can be seen from the figure that the proposed approach was able to reconstruct the frontal views very well regardless of the viewing angles.



Fig. 8. Some examples of reconstructed frontal faces of the same subject from the CMU PIE database. (a)–(d) Input images c05, c07, c29, and c37. (e)–(h) Reconstructed frontal views c05, c07, c29, and c37.

In order to evaluate the range of poses that can be handled by the method, we synthesized the frontal views of 2D face images generated from the USF 3D models at various viewing angles. The proposed approach can handle up to $\pm 30°$ in the pitch angle and $\pm 45°$ in the yaw angle. Figure 9 shows the synthesized frontal views for face images of the same person at four different poses. It can be seen from Figure 9(h) that the algorithm failed to reconstruct the frontal face image at the extreme pose. This is because most of the information on one half of the face is occluded due to the viewing angle. Another reason is that the extreme pose results in large image transformations that can not be handled by local warps of image patches.

C. Pose Invariant Face Recognition

As presented in above sections, it is more computationally efficient to classify whether a face image is frontal than to synthesize its frontal view (four seconds compared to two minutes). Thus, the frontal-view classifier is an important component of the proposed pose-invariant face recognition system. Before performing the recognition, the probe image was fed to the frontal-view classifier. If the image was classified as nonfrontal, it was transformed to the frontal view using the proposed algorithm. As a result, it is possible to perform recognition by combining our algorithm and any frontal face recognition technique. As we do not require the reference set to include an example of the person in the test image, the same two hundred *ba* frontal images from the FERET database were used as the training set for synthesizing frontal views in all the three face recognition experiments.

As in [8], if the face and both eyes cannot be detected using the cascade classifiers, a Failure to Acquire (FTA) has occurred. In this case, the frontal reconstruction is not carried out and the test image is not counted as a recognition error. The FTA rate is reported for each dataset in the recognition experiments below.

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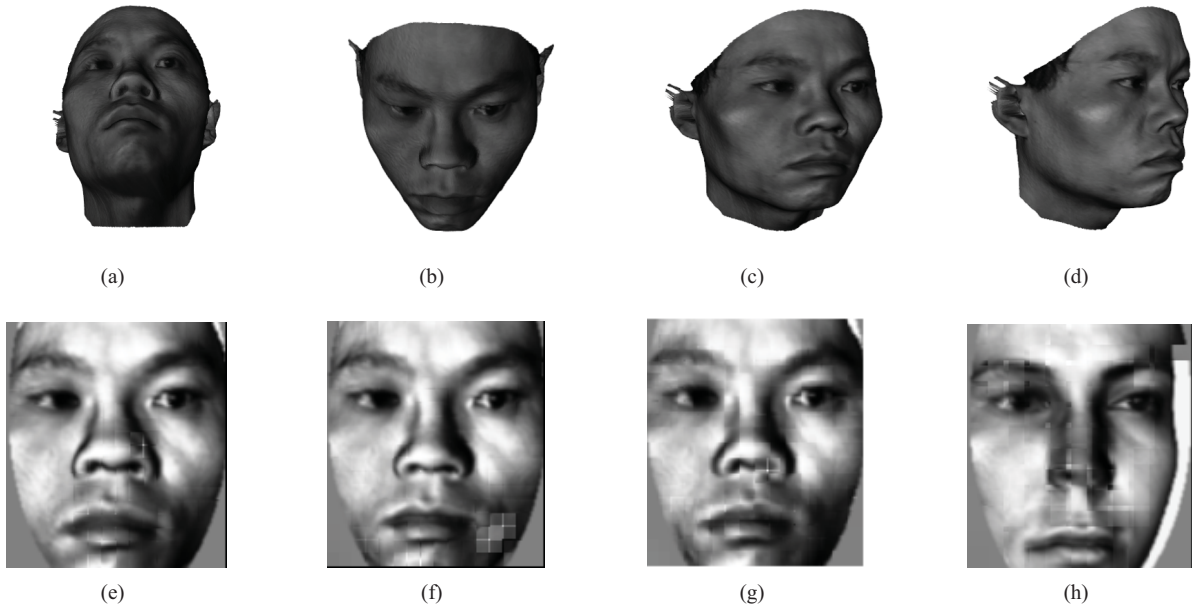


Fig. 9. Reconstructed frontal faces for input images at different poses from the USF 3D database. First row: 3-D face model of a person in the USF 3-D database at different viewing angles. (a) Pitch: +30°. (b) Pitch: −30°. (c) Yaw: −45°. (d) Yaw: −55°. Second row: 2-D frontal images synthesized using the proposed method. (e) Pitch: +30°. (f) Pitch: −30°. (g) Yaw: −45°. (h) Yaw: −55°.

TABLE II
RECOGNITION RATES OF DIFFERENT APPROACHES ON THE FERET DATABASE [24]. THE FRONTAL FACES *ba* WERE USED AS THE GALLERY IMAGES

Method	*bh* −40°	*bg* −25°	*bf* −15°	*be* +15°	*bd* +25°	*bc* +40°	**Average**
LGBP [53]	62.0%	91.0%	98.0%	96.0%	84.0%	51.0%	80.5%
LLR [16]	55.0%	89.5%	93.0%	89.0%	77.0%	53.0%	76.1%
PAN [54]	78.5%	91.5%	98.5%	97.0%	93.0%	81.5%	90.0%
3-D Pose norm. [8]	90.5%	98.0%	98.5%	97.5%	97.0%	91.9%	95.6%
Our approach	**91.0%**	**97.3%**	**98.0%**	**98.5%**	**96.5%**	**91.5%**	**95.5%**

TABLE III
RECOGNITION RATES OF DIFFERENT APPROACHES ON THE CMU-PIE DATABASE [25]. THE FRONTAL FACES *c*27 WERE USED AS THE GALLERY IMAGES

(a) 34 Faces							
Method	*c*11 −45°	*c*29 −22.5°	*c*05 +22.5°	*c*37 +45°	*c*07 up 22.5°	*c*09 down 22.5°	**Average**
ELF (Complex) [10]	78.0%	91.0%	93.0%	89.0%	95.0%	93.0%	89.8%
LLR [16]	89.7%	100.0%	98.5%	82.4%	98.5%	98.5%	94.0%
3ptSMD [12]	97.0%	100.0%	100.0%	100.0%	100.0%	100.0%	99.5%
Our approach	**97.0%**	**100.0%**	**100.0%**	**97.0%**	**97.1%**	**100.0%**	**98.5%**
(b) 68 Faces							
Method	*c*11 −45°	*c*29 −22.5°	*c*05 +22.5°	*c*37 +45°	*c*07 up 22.5°	*c*09 down 22.5°	**Average**
LGBP [53]	71.6%	87.9%	86.4%	75.8%	78.8%	93.9%	82.4%
3-D Pose norm. [8]	98.5%	100.0%	100.0%	97.0%	98.5%	100.0%	99.0%
Our approach	**97.0%**	**100.0%**	**100.0%**	**97.0%**	**98.5%**	**100.0%**	**98.8%**

In our experiments, the Local Gabor Binary Pattern (LGBP) [53] was selected as the face recognizer due to its effectiveness. In this method, a feature vector is formed by concatenating the histograms of all the local Gabor magnitude pattern maps over an input image. The histogram intersection is used as the similarity measurement in order to compare two feature vectors. More details about the application of the LGBP algorithm for face recognition can be found in [53].

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TABLE IV
RECOGNITION RATES OF DIFFERENT APPROACHES ON ONE HUNDRED AND THIRTY SEVEN SUBJECTS (SUBJECT ID 201 TO 346) WITH NEUTRAL EXPRESSIONS AND FRONTAL ILLUMINATION FROM THE MULTI-PIE DATABASE [26]. THE FRONTAL IMAGES FROM THE EARLIEST SESSION (POSE ID 051) WERE USED AS THE GALLERY IMAGES

Method	080_05 −45°	130_06 −30°	140_06 −15°	051_07 0°	051_08 +15°	041_08 +30°	190_08 +45°	**Average**
LGBP [53]	37.7%	62.5%	77.0%	92.6%	83.0%	59.2%	36.1%	64.0%
3-D Pose norm. [8]	74.1%	91.0%	95.7%	96.9%	95.7%	89.5%	74.8%	87.7%
Our approach	**86.3**%	**89.7**%	**91.7**%	**92.5**%	**91.0**%	**89.0**%	**85.7**%	**89.4**%

1) FERET Database: First, the recognition performance of our method on the FERET database is reported. We also compare our approach with the Local Gabor Binary Pattern (LGBP) [53], the Locally Linear Regression (LLR) method [16], the Piecewise Affine warping No stretch (PAN) approach [54], and a recent method based on 3D pose normalization [8]. The frontal faces *ba* were used as the gallery images. Table II shows the recognition rates of different methods for two hundred subjects at seven poses ranging from $-40°$ to $+40°$. It can be seen that the proposed approach outperformed the methods proposed in [16], [53], and [54]. The average rank-1 recognition rate of our algorithm was 95.5%, comparable to the result presented in [8] (95.6%). The FTA rate for the FERET dataset was 1.36%.

2) CMU-PIE Database: Next, we present the recognition results on the CMU PIE database. We compare our results with the ones presented in [12], [16], and [10] for thirty four faces using the same set-up where the gallery pose is frontal ($c27$) and the probe poses are $c05$, $c07$, $c09$, $c11$, $c29$, and $c37$. It can be seen from Table III that the proposed approach outperformed [16] and [10]. However, it was not as good as the stereo matching method in [12] (98.5% compared to 99.5%) which requires four landmark points. The proposed algorithm is also compared with the methods in [53] and [8] using all sixty eight faces in the CMU-PIE database. Table III shows that our recognition rate (98.8%) is better than the ones obtained by [53] (82.4%) and comparable to [8] (99.0%). For the this dataset, the FTA was 0.84%.

3) Multi-PIE Database: We also performed face recognition experiments on one hundred and thirty seven subjects (*Subject ID* 201 to 346) with neutral expressions and frontal illumination from the Multi-PIE database [26]. One hundred and thirty seven frontal images from the earliest session (*Pose ID* 051) were used as the gallery images. The probe set included the remaining images of both frontal (from other sessions) and nonfrontal views. The comparisons between our approach and the methods proposed in [53] and [8] on the Multi-PIE dataset are shown in Table IV. The average recognition rate achieved by our algorithm was better than the ones obtained by using the other two methods (89.4% compared to 64% and 87.7%). The FTA rate was 1.6%.

VI. CONCLUSION

In this paper, we presented a method for synthesizing the virtual frontal view from a nonfrontal face image. By dividing the input image into overlapping patches, a globally optimal set of local warps can be estimated to transform the patches to the frontal view. Each patch is aligned with images from a training database of frontal faces in order to obtain a set of possible warps for that node. It is worth noting that we do not require the training database to include the frontal images of the person in the test image. By using an extension of the LK algorithm that accounts for substantial illumination variations, the alignment parameters are calculated efficiently in the Fourier domain. The set of optimal warps is obtained by formulating the optimization problem as a discrete labeling algorithm using a discrete MRF and an efficient variant of the BP algorithm. The energy function of the MRF is also designed to handle illumination variations between different image patches. Furthermore, based on the sparsity of local SIFT descriptors, an efficient algorithm was also designed to classify whether the pose of the input face image is frontal or nonfrontal. Experimental results using the FERET, CMU PIE, and Multi-PIE databases show the effectiveness of the proposed approach.

In the future, we plan to investigate the possibility of synthesizing the probe image not only to the frontal pose, but also to other viewing angles. This will help the algorithm become more robust to large poses in the input images. A pyramidal implementation of the LK alignment algorithm can also be incorporated into the proposed approach in order to reduce the effect of patch size on the results [52].

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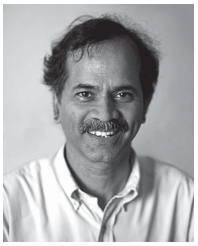


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